NLO single inclusive gluon production at mid rapidity: the finite terms

checked, DGLAP and running coupling separated, simplified, Gaussian projectile, colour-singlet form

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1 Summary

Input: your `Other_terms.tex` (in `input/`): all terms after JIMWLK ($\log \frac{k^+}{\Lambda}$) and BFKL ($\log \frac{\vee}{k^+}$); Groups I and III +c.c., Group II without.

1. **Check** (Sec. 2).
   - 25 of the 30 equations are identical, token by token, to the equations verified before: row derivations, BFKL extraction, the moduli of Group I Row III, the UV analysis, the $a \to 0$ limits.
   - The 5 that are new are Group I Row II part 1 and Group II Row IV with its definitions. They are correct, with one coefficient to fix: $\log(1 - \frac{\Lambda}{k^+})$ should have 4, not 1; it vanishes for $\Lambda \to 0$, so drop it.
   - Edits E1–E4 are listed in Sec. 2.3.
2. **DGLAP and running coupling** (Sec. 3).
   - The only poles left are in Group II Row IV and in Group I Row II part 1.
   - Group II Row IV gives DGLAP (the pole goes into $D_{h/g}(\mu_F)$) and the running coupling $\alpha_s \to \alpha_s(c_0/|y' - y|)$.
   - Group I Row II part 1 gives the constant $\frac{\alpha_s}{2\pi} N_c \big(\frac{67}{9} - \frac{\pi^2}{3}\big)$LO and two terms $\propto \log(1 - \frac{k^+}{\vee})$ (Q1, $V_\vee$).

3. **Leading power** (Sec. 4). All $k^+/\vee$ dependence outside the large logarithms is a power correction. At mid rapidity $\frac{k^+}{\vee} = \sqrt{\Lambda/\vee}$, as small as $\frac{\Lambda}{k^+}$, which is dropped everywhere. At leading power the result is consistent: Q1, the one pole that $D_{h/g}$ cannot absorb, is $O(\frac{k^+}{\vee})$. The exact-$\vee$ forms are kept in App. A.

4. **Simplified** (Sec. 5):
   - c.c. = 2Re;
   - every longitudinal integral becomes $\frac{1}{k^+}\times$ a dimensionless integral on $[0,1]$;
   - $2a$ and $2b$ of Group I Row V combine;
   - Group I Row V part 1 in closed form;
   - Group III in $\zeta = \frac{k^+}{p^+ + k^+}$ with the standard plus distribution;
   - rows with the same kernel combined.

5. **Gaussian projectile, colour singlet** (Secs. 6–10).
   - The three-ρ rows reduce to their two-ρ (merged-charge) parts.
   - Everything is $\int \mathcal{K}\mathcal{P}(u,v)\times$traces of adjoint Wilson lines.
   - Final result: Eq. (10.2).

6. **Open points** for the paper (App. B): the four-ρ finite term (none in your files), the large-$|x-y|$ behaviour of the R^i term, beyond-Gaussian projectiles.

2 Check of `Other_terms.tex`

equations	status	verified by
G1 II part 2 (R^i); G1 III (parts 1, 2, 3a, 3b, $\mathcal{A}_0$); G1 IV; G1 V (part 1, $2a$, $2b$ finite); G2 I (1a, 1b, 2a–2f); G3 I, II, III (1), III (2), V, VI	identical	row derivations (`rows_check`, `full_check`); BFKL extraction (`bfkl_final`); $\log\lvert\cdot\rvert$ by principal value (`neglog`, `Why_modulus`); UV finiteness (`uv_check`); limits (`limiting`); `checks.json`
G1 II part 1	new	Sec. 2.1
G2 IV and $\frac{d^3N_{\xi\mathrm{LO}}}{d^3k}$, $\frac{d^3N^y_{\xi\mathrm{LO}}}{d^3k}$, $\frac{d^3N_{\mathrm{LO}}}{d^3k}$	new	Sec. 3 and `g2iv_rc/compact`

2.1 Group I Row II part 1

Your equation has three pieces:

1. **The β_0 terms are gone, correctly.** The $\frac{11}{3}$ of this row and the $-\frac{11}{6}$ of G1 V $2b$ UV cancel exactly: written in the form of this row, the $2b$ UV term is $-\frac{11}{3}$. So both the $\frac{11}{3}$ line and $2b$ UV can be dropped.

2. **Pole** $\times\big[\log(1-\frac{\Lambda}{k^+}) + \log(1-\frac{k^+}{\vee})\big]$.
   - $\log(1-\frac{k^+}{\vee})$ is right (Q1, Sec. 3).
   - $\log(1-\frac{\Lambda}{k^+})$ comes from $2b$ UV, $\frac{g^4N_c}{16\pi^4}\,2\log(1-\frac{\Lambda}{k^+})$. In the form of this row ($\frac{g^4N_c}{32\pi^4}$, colour $[U(w)-U(x)]$) its coefficient is $4\log(1-\frac{\Lambda}{k^+})$, not 1. It vanishes for $\Lambda\to 0$: drop it.

3. **Finite:** $-\frac{67}{9}+\frac{\pi^2}{3}+\log^2(1-a)+\log(1-a)\log\frac{\vee}{k^+} = -\frac{67}{9}+\frac{\pi^2}{3}+\log(1-a)\log\big(\frac{\vee}{k^+}-1\big)$, $a\equiv\frac{k^+}{\vee}$. Correct.

2.2 Group II Row IV

Correct, with your definitions of $\frac{d^3N_{\xi\mathrm{LO}}}{d^3k}$, $\frac{d^3N^y_{\xi\mathrm{LO}}}{d^3k}$, $\frac{d^3N_{\mathrm{LO}}}{d^3k}$. Three points:

- $\mathcal{P}_{gg}$ **must be defined** (E4): $\mathcal{P}_{gg} = P_{gg} = 2N_c\big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\big] + \frac{\beta_0}{2}\delta(1-\xi)$. The plus distribution is the standard one on $[0,1]$, used on $[a,1]$: $\int_a^1 \frac{g(\xi)}{(1-\xi)_+} = \int_a^1 \frac{g(\xi)-g(1)}{1-\xi} + g(1)\log(1-a)$. This $\log(1-a)$ is your former separate line.
- **The weights belong inside the integrals.** The brackets $\big(\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2}{4}\big)$ and $\log(\dots)$ depend on $y'-y$, so they stand inside $\int_{x,y,x',y'}$ of $\frac{d^3N_{\xi\mathrm{LO}}}{d^3k}$, $\frac{d^3N^y_{\xi\mathrm{LO}}}{d^3k}$, $\frac{d^3N_{\mathrm{LO}}}{d^3k}$.
- **The sign of the β_0 line is $+$**, as you now have.

Also correct: the $-\frac{2}{\epsilon}$ form of the bracket ($d_\perp = 2-\epsilon$: $-\frac{d_\perp}{\epsilon} - \dots - 1 = -\frac{2}{\epsilon} - \frac{d_\perp}{2}\ell_\Delta$), $(1-a)H_4$, and $d_\perp\to 2$ in the linear-polarization line.

2.3 Edits

E1 G1 II part 1: $\log(1-\frac{\Lambda}{k^+}) \to 4\log(1-\frac{\Lambda}{k^+})$, or drop it ($\Lambda \to 0$). At leading power (Sec. 4) the $\log(1-a)$ terms also go.

E2 G1 V $2b$ finite: $\int_\Lambda^{k^+-\Lambda} \to \int_0^{k^+}$, $\int_{k^+-\Lambda}^{\Lambda} \to \int_{k^+}^{0}$. The integrands are regular at the ends.

E3 Group III at leading power: $\int_0^{\vee-k^+} \to \int_0^{\infty}$. The integrands fall like $1/p^{+2}$ (`limiting/largep.py`).

E4 Define $\mathcal{P}_{gg}$ as above.

3 Poles: DGLAP and running coupling

Notation: $\ell_\Delta(\mu) \equiv 2\gamma_E + \log\frac{(y'-y)^2\mu^2}{4} = \log\frac{(y'-y)^2\mu^2}{c_0^2}$, $c_0 = 2e^{-\gamma_E}$, $\beta_0 = \frac{11}{3}N_c$ ($n_f = 0$), $\mathrm{LO}_\xi[f]$ is your $\frac{d^3N_{\xi\mathrm{LO}}}{d^3k}$ with the weight f inside the integral, $\mathrm{LO}_\xi^y[f]$ likewise, $\mathrm{LO} \equiv \mathrm{LO}_{\xi=1}[1]$.

source	pole and μ	goes into
G2 IV, $\mathcal{P}_{gg}$ line	$\frac{\alpha_s}{2\pi}\int_a^1 \frac{d\xi}{\xi^2} P_{gg}\mathrm{LO}_\xi\big[-\frac{2}{\epsilon} - \ell_\Delta(\mu)\big]$	$D_{h/g}(\mu_F)$: DGLAP
G2 IV, β_0 line + coupling counterterm of LO	$\frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}\big[\frac{2}{\epsilon} + \ell_\Delta(\mu)\big] - \frac{\alpha_s\beta_0}{4\pi}\frac{2}{\epsilon}\mathrm{LO}$	running coupling
G2 IV, LO_ξ^y line	none: the μ part multiplies $\int_{y'} e^{-ik\cdot(y'-y)/\xi} \propto \delta^2(\mathbf{k})$	small-ξ subtraction
G1 II part 1 +c.c.	$\frac{\alpha_s}{2\pi}\big(-\frac{2}{\epsilon} - \ell\big)\big(-N_c\log(1-a)\big)\mathrm{LO}$	Q1, $O(a)$

DGLAP. In $\frac{d^3N_h}{d^3k} = \int_{k^+/\vee}^1 \frac{d\zeta}{\zeta^3} D_{h/g}(\zeta)\frac{d^3N_g}{d^3k}\big(\frac{k^+}{\zeta}, \frac{\mathbf{k}}{\zeta}\big)$ the gluon has $\frac{k^+}{\zeta} \le \vee$, so its ξ integral starts at $\frac{k^+}{\zeta\vee}$.

- The pole region $a \le \zeta\xi \le \zeta \le 1$ is exactly that of the $\overline{\mathrm{MS}}$ counterterm $\int_\zeta^1 \frac{dz}{z} P_{gg}(z)D(\zeta/z)$, with the standard plus distribution.
- So $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon})P_{gg} \otimes \mathrm{LO}$ is absorbed completely, $\log(1-a)$ endpoint term included. What is left is

$$\left.\frac{d^3N}{d^3k}\right|_{\mathrm{DGLAP}} = -\frac{\alpha_s}{2\pi}\int_a^1 \frac{d\xi}{\xi^2}\Big\{P_{gg}(\xi)\,\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big] - 2N_c\frac{1-\xi}{\xi}\,\mathrm{LO}_\xi^y\big[\ell_\Delta(\mu_F)\big]\Big\}. \tag{3.1}$$

- The LO_ξ^y term removes the $\log\frac{\vee}{k^+}$ that the $\frac{1-\xi}{\xi}$ part would otherwise produce at $\xi \to a$ (it is in BFKL); the two must stay under one integral.

Running coupling.

$$\left.\frac{d^3N}{d^3k}\right|_{\mathrm{RC}} = \frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}\Big[\log\frac{(y'-y)^2\mu_R^2}{c_0^2}\Big], \qquad \mathrm{LO}\big|_{\alpha_s(\mu_R)} + \left.\frac{d^3N}{d^3k}\right|_{\mathrm{RC}} = \mathrm{LO}\big|_{\alpha_s\to\alpha_s(c_0/|y'-y|)} + O(\alpha_s^3). \tag{3.2}$$

The constant of G1 II part 1. With the c.c.:

$$\frac{\alpha_s}{2\pi}N_c\Big(\frac{67}{9} - \frac{\pi^2}{3}\Big)\mathrm{LO} = \frac{\alpha_s}{2\pi}\,2K\,\mathrm{LO}, \qquad K = N_c\Big(\frac{67}{18} - \frac{\pi^2}{6}\Big), \tag{3.3}$$

K being the constant of the two-loop cusp anomalous dimension (CMW). Correction to the earlier `Limiting_terms.pdf` (now fixed there): it said that it is absorbed by $\alpha_s \to \alpha_s(1+\frac{\alpha_s}{2\pi}K)$. Since LO $\propto \alpha_s$, the coefficient here is $2K$, i.e. $\alpha_s \to \alpha_s(1+\frac{\alpha_s}{\pi}K)$, twice the CMW shift. Keep it as the explicit constant (3.3). Whether part of it combines with constants hidden in the finite part of $2b$ is not decided by these terms.

Q1 and $V_\vee$ (from the cut $\xi \ge a$ in the virtual term of G1 II):

$$\left.\frac{d^3N}{d^3k}\right|_{\mathrm{Q1}} = \frac{\alpha_s}{2\pi}N_c\log(1-a)\,\mathrm{LO}\Big[\frac{2}{\epsilon} + \ell\Big], \qquad \left.\frac{d^3N}{d^3k}\right|_{V_\vee} = -\frac{\alpha_s}{2\pi}N_c\log(1-a)\log\Big(\frac{1}{a}-1\Big)\mathrm{LO}, \tag{3.4}$$

where ℓ is the half sum of the logarithms at the separations $x-w$ and $x'-w'$ (the two orderings). Q1 is a pole that neither counterterm removes; both are $O(a\log a)$.

4 Leading power in $a = k^+/\vee$

- **The equations converge to their $a \to 0$ forms like a.** Taking the whole of every equation (`limiting/limcheck.py`), the maximum relative difference between the exact-a and $a \to 0$ forms is 0.35, 3.5×10^{-3}, 3.5×10^{-5}, 3.5×10^{-7} at $a = 10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}$.
- **Mid rapidity.** There $k^+ = \sqrt{\vee\Lambda}$, so $a = \sqrt{\Lambda/\vee} = \frac{\Lambda}{k^+}$. The terms $O(\frac{\Lambda}{k^+})$ are dropped everywhere ($\Lambda \to 0$), so keeping $O(a)$ would be inconsistent.
- **Q1 forces it.** At finite a the pole Q1 is not absorbed. At leading power the result has no uncancelled pole. (Alternatively, integrate the virtual ξ over $[0,1]$, as AABKL do; then Q1 and $V_\vee$ are absent at any a.)

Therefore the paper result is at leading power in a and in $\frac{\Lambda}{k^+}$, with the large logarithms $\log\frac{k^+}{\Lambda}$ and $\log\frac{\vee}{k^+}$ kept exactly. App. A lists where a enters if you want to quote exact-$\vee$ forms.

5 Simplified finite terms (leading power)

Every expression below is checked in `limiting/checks.json` (Ein closed forms, ζ measures and plus prescription, phase identities, the $q^+ \to \xi$ map, the closed-form p^+ integral).

5.1 The c.c.

The charges are hermitian and the adjoint Wilson lines real, so the c.c. of a term is: $i \to -i$, $e^{-ik\cdot(\cdot)} \to e^{+ik\cdot(\cdot)}$, and the order of the ρ's reversed. Its expectation value in the projectile is the complex conjugate. Hence, for Groups I and III,

$$\big\langle\, d^3N_{\rm row} + \text{c.c.}\,\big\rangle = 2\,\mathrm{Re}\,\big\langle d^3N_{\rm row}\big\rangle, \qquad \text{and for the rows with } i\,f\text{:}\quad 2\,\mathrm{Re}\,\langle iX\rangle = -2\,\mathrm{Im}\,\langle X\rangle. \tag{5.1}$$

Group II is self-conjugate ($1a \leftrightarrow 1b$; in part 2 every term comes with its $(x,w) \leftrightarrow (x',w')$ partner).

5.2 Group I

Row II, part 1 (CMW constant). Relative to LO the prefactor is $\frac{\alpha_s N_c}{4\pi}$, the c.c. doubles it, and $[U(x')-U(w')][U(w)-U(x)] = -[U(w')-U(x')][U(w)-U(x)]$:

$$d^3N_{\mathrm{G1\,II(1)}} + \text{c.c.} = \frac{\alpha_s N_c}{2\pi}\Big(\frac{67}{9} - \frac{\pi^2}{3}\Big)\mathrm{LO} = \frac{\alpha_s}{2\pi}\,2K\,\mathrm{LO}, \qquad K = N_c\Big(\frac{67}{18} - \frac{\pi^2}{6}\Big). \tag{5.2}$$

Since LO $\propto \alpha_s$, absorbing it into the coupling of LO needs $\alpha_s \to \alpha_s\big(1 + \frac{\alpha_s}{\pi}K\big)$, twice the CMW shift; keep it as an explicit constant.

Row II, part 2 (R^i), Rows III, IV: as in your file (no longitudinal integral left), with E1 and +c.c.

Row V, first part: one ξ integral and closed form. With $\xi = \frac{p^+}{k^+}$ in line 1 and $\xi = 1 - \frac{q^+}{k^+}$ in line 2 both phases become $e^{-ik\cdot(w'-(1-\xi)z-\xi x)}$, and $\frac{1}{k^+}\int_0^{k^+} dq^+\big[\frac{1}{q^+}\big]_+ = \int_0^1 d\xi\big[\frac{1}{1-\xi}\big]_+$:

$$\begin{aligned}\frac{d^3N_{\mathrm{V(1)}}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{ig^4}{32\pi^5}\frac{1}{k^+}\int_0^1 d\xi \int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-(1-\xi)z-\xi x)}\,\frac{(x'-w')^i(z-x)^m(y-z)^k(y'-x)^j}{(x'-w')^2(z-x)^2(y-z)^2(y'-x)^2}\\ &\quad\times\Big[\delta_{im}\delta_{jk} - \Big[\frac{1}{\xi}\Big]_+\delta_{jm}\delta_{ik} - \Big[\frac{1}{1-\xi}\Big]_+\delta_{ij}\delta_{mk}\Big]\,\mathcal{C}_{\mathrm{V(1)}} \qquad (5.3)\\ &= \frac{1}{(2\pi)^3}\frac{ig^4}{32\pi^5}\frac{1}{k^+}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')^i(z-x)^m(y-z)^k(y'-x)^j}{(x'-w')^2(z-x)^2(y-z)^2(y'-x)^2}\\ &\quad\times\Big[\frac{1-e^{-i\alpha}}{i\alpha}\,\delta_{im}\delta_{jk} + \mathrm{Ein}(i\alpha)\,\delta_{jm}\delta_{ik} + e^{-i\alpha}\mathrm{Ein}(-i\alpha)\,\delta_{ij}\delta_{mk}\Big]\,\mathcal{C}_{\mathrm{V(1)}}, \qquad \alpha \equiv k\cdot(z-x), \qquad (5.4)\end{aligned}$$

$\mathrm{Ein}(s) = \int_0^1 \frac{1-e^{-st}}{t}dt = \sum_{n\ge1}\frac{(-1)^{n+1}s^n}{n\,n!} = \gamma_E + \log s + E_1(s)$ (entire), and $\mathcal{C}_{\mathrm{V(1)}} = \big[U^{ab'}(x') - U^{ab'}(w')\big]\rho^{b'}(x')\big[f^{adc}U^{db}(x)U^{ce}(z)\{\rho^e(y),\rho^b(y')\} - f^{abc}U^{cd}(y)U^{be}(y')\{\rho^d(y),\rho^e(y')\}\big]$.

Row V, second part: $2a$ and $2b$ finite combine. With $\xi = \frac{p^+}{k^+}$ in the p^+ lines and $\xi = 1 - \frac{q^+}{k^+}$ in the q^+ lines, all lines of $2a$ and of $2b$ finite have the same phase, the same denominator k^+D_ξ,

$$D_\xi \equiv \xi\,(y-x)^2 + (1-\xi)\,(y-z)^2, \tag{5.5}$$

and the same kernel, Eq. (5.7); they differ only by the colour factor. Therefore (E2 used, $d_\perp \to 2$):

$$\frac{d^3N_{2a}}{d^3k} + \frac{d^3N_{2b\,\mathrm{fin}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\int_0^1 d\xi\int_{x,y,z,x',w'}\frac{e^{-ik\cdot(w'-\xi x-(1-\xi)z)}}{D_\xi}\,\frac{(x'-w')^i(z-x)^m}{(x'-w')^2(z-x)^2}\,\mathcal{B}^{im}(\xi)$$

$$\times \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c U^{af}(y) - f^{adc}f^{fbe}U^{db}(x)U^{ce}(z)\Big]\rho^{b'}(x')\rho^{f}(y)$$
$$+ \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\frac{11}{6}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}$$
$$\times \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c U^{af}(y) - N_c U^{af}(z)\Big]\rho^{b'}(x')\rho^{f}(y), \tag{5.6}$$

$$\mathcal{B}^{im}(\xi) = \xi(1-\xi)\,\frac{\delta^{im}\big[(y-x)^2-(y-z)^2\big]}{(x-z)^2} + (1-\xi)\,\delta^{im}\Big[\frac{(y-z)\cdot(x-z)}{(x-z)^2} + \frac{(y-x)\cdot(y-z)}{2(y-x)^2} - \frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Big]$$
$$+\,\xi\,\delta^{im}\Big[\frac{(y-x)\cdot(x-z)}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{2(y-z)^2} - \frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Big] - \Big(\Big[\frac{1}{\xi}\Big]_+ - 1\Big)S^{im} - \Big(\Big[\frac{1}{1-\xi}\Big]_+ - 1\Big)T^{im} - S'^{im}, \tag{5.7}$$

$$S^{im} = \frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2}, \qquad T^{im} = \frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2},$$
$$S'^{im} = \frac{(y-x)^m(x-z)^i}{(x-z)^2} - \frac{(y-x)^m(y-z)^i}{2(y-z)^2} + \frac{(y-z)^m(x-z)^i}{(x-z)^2} + \frac{(y-x)^i(y-z)^m}{2(y-x)^2}.$$

The $2a$ colour factor vanishes at $z=x$ ($-f^{adc}f^{fbe}U^{db}(x)U^{ce}(x) + N_cU^{af}(x) = 0$, checked), so the $x\to z$ behaviour of the sum is that of $2b$, removed by the $\frac{11}{6}$ line: Eq. (5.6) is finite as written. The weights $\xi(1-\xi)$, $1-\xi$, ξ, $\big[\frac{1}{\xi}\big]_+$, $\big[\frac{1}{1-\xi}\big]_+$ are those of the virtual $g\to gg$ kernel, and $\int_0^1 d\xi\,[\xi(1-\xi)-2] = -\frac{11}{6}$ (Sec. 11).

5.3 Group II

Row I: as in your file (no c.c.; no longitudinal integral; $d_\perp \to 2$ in $2f$, three lines together).

Row IV: DGLAP + running coupling + finite. Your equation, with $\mathcal{P}_{gg} = 2N_c\big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\big] + \frac{\beta_0}{2}\delta(1-\xi)$ on $[0,1]$ and the brackets inside $\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}$ (`g2iv_rc/compact`), gives

$$\text{DGLAP:}\quad -\frac{\alpha_s}{2\pi}\int_0^1\frac{d\xi}{\xi^2}\Big\{P_{gg}(\xi)\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big] - 2N_c\frac{1-\xi}{\xi}\mathrm{LO}^y_\xi\big[\ell_\Delta(\mu_F)\big]\Big\}\quad(\text{pole in } D_{h/g}(\mu_F)), \tag{5.8}$$

$$\text{running coupling:}\quad \frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\big[\ell_\Delta(\mu_R)\big] \quad\Rightarrow\quad \mathrm{LO}\big|_{\alpha_s\to\alpha_s(c_0/|y'-y|)}, \tag{5.9}$$

$$\text{finite:}\quad -H_4 + \frac{\alpha_s}{2\pi}\,2N_c\int_0^1 d\xi\,\frac{1-\xi}{\xi}\,\mathrm{LO}^{\mathrm{lp}}_\xi\Big|_{d_\perp=2}. \tag{5.10}$$

5.4 Group III: $\zeta = \frac{k^+}{p^+ + k^+}$

With $p^+ = k^+\frac{1-\zeta}{\zeta}$ ($p^+: 0\to\infty$ is $\zeta: 1\to 0$; E3 used):

$$\frac{dp^+}{p^+ + k^+} = \frac{d\zeta}{\zeta}, \qquad \Big[\frac{k^+}{p^+(p^+ + k^+)}\Big]_+ dp^+ = \Big[\frac{1}{1-\zeta}\Big]_+ d\zeta, \qquad \frac{dp^+}{k^+} = \frac{d\zeta}{\zeta^2}, \tag{5.11}$$

$$e^{-ik\cdot(y'-w)}e^{-ik\cdot\frac{p^+}{k^+}(y'-z)} = e^{-i\frac{k}{\zeta}\cdot\left(y'-\zeta w-(1-\zeta)z\right)} \tag{5.12}$$

(your plus prescription, subtraction at $p^+=0$, is the standard one at $\zeta=1$; all checked numerically). Rows III part 1, V, VI then read, with $P_{\mathrm{III}(1)} = \frac{ig^4 f^{c'd'a}}{16\pi^5}$, $P_{\mathrm{V}} = \frac{ig^4 f^{c'ca}}{8\pi^5}$, $P_{\mathrm{VI}} = \frac{ig^4 f^{c'd'a}}{8\pi^5}$,

$$\frac{d^3N_{\mathrm{G3\,X}}}{d^3k} = \frac{1}{(2\pi)^3}P_{\mathrm{X}}\frac{1}{k^+}\int_0^1 d\zeta\int e^{-i\frac{k}{\zeta}\cdot(y'-\zeta w-(1-\zeta)z)}\Big\{\Big[\frac{1}{\zeta}(y'-z)\cdot(x'-y')\,(y-w)\cdot(x-z)$$
$$-\Big(\Big[\frac{1}{1-\zeta}\Big]_+ + \frac{1}{\zeta}\Big)(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)\Big]\frac{\mathcal{C}_{\mathrm{X}}\,\mathcal{C}'_{\mathrm{X}}}{(y'-z)^2(x'-y')^2(y-w)^2(x-z)^2}$$
$$-\frac{1}{\zeta^2}\frac{(y'-z)\cdot(y-w)}{(y'-z)^2(y-w)^2(x-z)^2}\Big[\frac{(x-z)\cdot(x'-y')}{(x'-y')^2}\mathcal{C}_{\mathrm{X}} - \frac{(x-z)\cdot(x'-z)}{(x'-z)^2}\mathcal{C}_{\mathrm{X}}\big|_{U(y')\to U(z)}\Big]\mathcal{C}'_{\mathrm{X}}\Big\}, \tag{5.13}$$

with the colour factors of your file: $\mathcal{C}_{\mathrm{III}(1)} = \mathcal{C}_{\mathrm{VI}} = \big[U^{e'c'}(y')U^{d'b}(z) - U^{bd'}(z)U^{c'e'}(x')\big]\rho^{e'}(x')$, $\mathcal{C}_{\mathrm{V}} = \big[U^{e'c'}(y') - U^{c'e'}(x')\big]\rho^{e'}(x')$; $\mathcal{C}'_{\mathrm{III}(1)} = U^{ac}(w)\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)U^{ad}(y)U^{ce}(x)\{\rho^d(y),\rho^e(x)\}$, $\mathcal{C}'_{\mathrm{V}} = \big[U^{ce}(x)U^{ad}(y) - U^{ad}(w)U^{ce}(x)\big]\rho^e(x)\rho^d(y)$, $\mathcal{C}'_{\mathrm{VI}} = U^{bc}(z)U^{ad}(y)U^{ce}(x)\rho^d(y)\rho^e(x) - U^{ad}(y)\rho^d(y)\rho^b(x)$. The second line is the $y'\to z$ subtraction; it is what makes $\zeta\to 0$ ($p^+\to\infty$) integrable.

Row III, part 2. With $A \equiv (x-z)^2$, $B \equiv (x-w)^2$, $D_\zeta \equiv (1-\zeta)A + \zeta B$, $\frac{dp^+}{p^+A+k^+B} = \frac{d\zeta}{\zeta D_\zeta}$:

$$\begin{aligned}
\frac{d^3N_{\mathrm{III}(2)}}{d^3k} = {}& \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_0^1 d\zeta \int_{w,x',y',z,x} e^{-i\frac{k}{\zeta}\cdot(y'-\zeta w-(1-\zeta)z)}\frac{(y'-z)^m(x'-y')^{k'}}{(y'-z)^2(x'-y')^2}\Big\{\frac{\delta_{k'm}}{D_\zeta}\Big[\frac{(1-\zeta)(A-B)}{(z-w)^2}+\frac{1-\zeta}{\zeta}\,b+c\Big] \\
& - \frac{1}{\zeta D_\zeta}\,e^{k'm} - \Big(\Big[\frac{1}{1-\zeta}\Big]_+\frac{\zeta}{D_\zeta}+\frac{1}{D_\zeta}\Big)d^{k'm}\Big\}\,\mathcal{C}_{\mathrm{VI}}\,\mathcal{C}'' \\
& - \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_0^1 d\zeta\,\frac{1-\zeta}{\zeta^2}\int e^{-i\frac{k}{\zeta}\cdot(y'-\zeta w-(1-\zeta)z)}\frac{(y'-z)^m}{(y'-z)^2}\,\frac{t^{k'm}}{D_\zeta}\Big[\frac{(x'-y')^{k'}}{(x'-y')^2}\mathcal{C}_{\mathrm{VI}} - \frac{(x'-z)^{k'}}{(x'-z)^2}\mathcal{C}_{\mathrm{VI}}\big|_{U(y')\to U(z)}\Big]\mathcal{C}'' \\
& + \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\frac{(w'-z)^m(x'-z)^{k'}}{(w'-z)^2(x'-z)^2}\,\frac{(x-w)^2\log\frac{(x-w)^2}{(x-z)^2}}{(x-z)^2\big[(x-w)^2-(x-z)^2\big]}\,t^{k'm}\,\mathcal{C}_{\mathrm{VI}}\big|_{U(y')\to U(z)}\,\mathcal{C}'',
\end{aligned} \tag{5.14}$$

where $\mathcal{C}'' = f^{cbd}U^{ac}(w)\rho^d(x) - f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)$ (times $f^{c'd'a}$ in $\mathcal{C}_{\mathrm{VI}}$), and $b, c, d^{k'm}, e^{k'm}, t^{k'm}$ are your brackets: $b = \frac{(x-z)\cdot(z-w)}{(z-w)^2} - \frac{(x-z)\cdot(x-w)}{2(x-w)^2} - \frac{A-B}{2(z-w)^2}$, $c = \frac{(x-w)\cdot(z-w)}{(z-w)^2} + \frac{(x-z)\cdot(x-w)}{2(x-z)^2} - \frac{A-B}{2(z-w)^2}$, $d^{k'm} = \frac{(x-w)^{k'}(z-w)^m}{(z-w)^2} + \frac{(x-z)^m(x-w)^{k'}}{2(x-z)^2}$, $e^{k'm} = \frac{(x-z)^m(z-w)^{k'}}{(z-w)^2} - \frac{(x-z)^m(x-w)^{k'}}{2(x-w)^2} + \frac{(x-w)^m(z-w)^{k'}}{(z-w)^2} + \frac{(x-z)^{k'}(x-w)^m}{2(x-z)^2}$, $t^{k'm} = \frac{(x-z)^{k'}(z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'}(x-w)^m}{2(x-w)^2}$. A plus distribution acts on everything to its right, phase included ($\int_0^1 d\zeta\big[\frac{1}{1-\zeta}\big]_+\Phi(\zeta) = \int_0^1 d\zeta\frac{\Phi(\zeta)-\Phi(1)}{1-\zeta}$). The last line is your third line with its p^+ integral done:

$$\int_0^\infty dp^+\frac{B}{A\,(p^++k^+)(p^+A+k^+B)} = \frac{B\log\frac{B}{A}}{k^+A(B-A)} \qquad \text{(regular at } A=B\text{; checked to } 10^{-15}\text{)}. \tag{5.15}$$

6 Rules

Reduction. Write a three-ρ colour operator with your charge order, $\mathcal{O} = T^{a_1a_2a_3}\rho^{a_1}(u_1)\rho^{a_2}(u_2)\rho^{a_3}(u_3)$ ($\{B,C\}$ written out). Up to terms with one ρ, which vanish for $\langle\rho\rangle = 0$,

$$\mathcal{O} = S(\mathcal{O}) + \mathcal{O}^{(2)}, \qquad \mathcal{O}^{(2)} = \frac{1}{2}\sum_{i<j}\mathcal{O}\Big|_{\rho^{a_i}(u_i)\rho^{a_j}(u_j)\to[\rho^{a_i}(u_i),\rho^{a_j}(u_j)]}, \qquad [\rho^a(u),\rho^b(v)] = if^{abc}\rho^c(u)\,\delta^{(2)}(u-v). \tag{6.1}$$

($ABC - \sigma(ABC)$ is the sum of the commutators of the pairs that σ inverts; averaged over the six orders each pair is inverted half of the time.) S is the fully symmetric part. For a colour singlet $\langle S(\rho\rho\rho)\rangle \propto d^{abc}\langle d\rho\rho\rho\rangle$, $f^{abc}\langle fS(\rho\rho\rho)\rangle$; both vanish for a Gaussian.

Kernels. The $\delta^{(2)}(u_i-u_j)$ removes the integral over u_j. The kernel of the row is evaluated at $u_j = u_i$: written $\mathcal{K}\big|_{u_j\to u_i}$. The third charge u_k stays. So

$$\Big\langle\frac{d^3N_{\mathrm{row}}}{d^3k}\Big\rangle = \frac{1}{(2\pi)^3}\int\sum_{\text{merges}}\mathcal{K}_{\mathrm{row}}\big|_{u_j\to u_i}\,\mathcal{P}(u_i,u_k)\,C^{\mathrm{row}}_{u_j\to u_i}, \tag{6.2}$$

with C the projected colour factor of that merge (merges not listed are zero, e.g. the pair inside an anticommutator).

Index placement. As in `Finite_terms_singlet.pdf`, Sec. 2.2. Group I as written; Groups II and III with the general-U placement of correction M4 (identical to yours for $U = U^T$).

c.c. After the projection every trace is real (U real; two T^e's give a real product) and $\mathcal{P}$ is real and symmetric. So for Groups I and III

$$\mathrm{row} + \mathrm{c.c.} = 2\,\mathrm{Re}\,(\mathrm{row}). \tag{6.3}$$

In the $\mathcal{K}_{\mathrm{IV}}$ family the Group II Row I part 1 term enters with its block A; block B ($1b$) is its complex conjugate, so 2Re covers it too.

7 Kernels ($d_\perp = 2$, leading power)

All kernels include their prefactor, phase and longitudinal integral; the transverse integrals are those of the row.

$$\begin{aligned}
\mathcal{K}_{\mathrm{II}} = {}& -\frac{g^4}{64\pi^6k^+}\,e^{-ik\cdot(w'-w)}\int d^2p\,d^2q\;e^{iq\cdot(w-y)}e^{ip\cdot(y-x)}\frac{(x'-w')^i\,R^i(\mathbf{q},\mathbf{p})}{(x'-w')^2\,\mathbf{p}^2\mathbf{q}^2}, \\
R^i = {}& -2q^i\log\frac{p^2}{q^2} + \frac{(2p^2-2q\cdot p+q^2)q^i+q^2p^i}{(q-p)^2}\log\frac{q^2}{(q-p)^2}
\end{aligned} \tag{7.1}$$

$$\mathcal{K}_{\mathrm{III}} = \text{the sum of the four parts of your Row III (unchanged) without the colour bracket,} \tag{7.2}$$

$$\mathcal{K}_{\rm IV} = \frac{ig^4}{8\pi^5 k^+} e^{-ik\cdot(w'-w)} \frac{(x'-w')^i (y-z)^j}{(x'-w')^2(y-z)^2} \log\frac{(x-z)^2}{(x-w)^2} \Big[\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2} - \frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2} - \frac{(x-z)^j(x-w)^i}{(x-z)^2(x-w)^2} + \frac{\delta^{ij}}{2(z-w)^2}\Big], \tag{7.3}$$

$$\mathcal{K}_{\rm V1} = \frac{ig^4}{32\pi^5 k^+} \int_0^1 d\xi\, e^{-ik\cdot(w'-(1-\xi)z-\xi x)} \frac{(x'-w')^i (z-x)^m (y-z)^k (y'-x)^j}{(x'-w')^2(z-x)^2(y-z)^2(y'-x)^2} \Big[\delta_{im}\delta_{jk} - \Big[\frac{1}{\xi}\Big]_+ \delta_{jm}\delta_{ik} - \Big[\frac{1}{1-\xi}\Big]_+ \delta_{ij}\delta_{mk}\Big], \tag{7.4}$$

$$\mathcal{K}_{C} = \frac{ig^4}{16\pi^5 k^+} \int_0^1 d\zeta\, e^{-i\frac{k}{\zeta}\cdot(y'-\zeta w-(1-\zeta)z)} \times \frac{\frac{1}{\zeta}(y'-z)\cdot(x'-y')(y-w)\cdot(x-z) - \big(\big[\frac{1}{1-\zeta}\big]_+ + \frac{1}{\zeta}\big)(y'-z)\cdot(x-z)(x'-y')\cdot(y-w)}{(y'-z)^2(x'-y')^2(y-w)^2(x-z)^2}, \tag{7.5}$$

$$\mathcal{K}_{C2} = \frac{ig^4}{16\pi^5 k^+} \int_0^1 \frac{d\zeta}{\zeta^2}\, e^{-i\frac{k}{\zeta}\cdot(y'-\zeta w-(1-\zeta)z)} \frac{(y'-z)\cdot(y-w)\,(x-z)\cdot(x'-y')}{(y'-z)^2(y-w)^2(x-z)^2(x'-y')^2}, \qquad \mathcal{K}_{C2z} = \mathcal{K}_{C2}\Big|_{\frac{(x'-y')}{(x'-y')^2}\to\frac{(x'-z)}{(x'-z)^2}}. \tag{7.6}$$

(In $\mathcal{K}_{\rm II}$ the second momentum is written q: your $\int d^2k$ uses the letter of the measured gluon.) Charges: Group I Rows II, III, IV and Group III: (x', x, y); Group I Row V part 1: (x', y, y').

- $\mathcal{K}_{\rm IV}$ **family:** Group I Row IV + Group III Rows I, II + Group II Row I part 1 $= \frac{1}{(2\pi)^3}\int \mathcal{K}_{\rm IV}\big(\mathcal{O}_{\rm IV} - \mathcal{O}_{\rm G3I} - \mathcal{O}_{\rm G3II} - \frac{1}{2}\mathcal{O}_A\big)$ (+c.c.).
- $\mathcal{K}_C$ **family:** Group III Rows III part 1, V, VI $= \frac{1}{(2\pi)^3}\int \big[\mathcal{K}_C\mathcal{O}_C - \mathcal{K}_{C2}\mathcal{O}_C + \mathcal{K}_{C2z}\mathcal{O}^z_C\big]$ (+c.c.), where:
  - $\mathcal{O}_C = \mathcal{O}_{\rm III(1)} + 2\mathcal{O}_{\rm V} + 2\mathcal{O}_{\rm VI}$ (Eq. (5.13), weights $\frac{P_X}{P_{\rm III(1)}}$);
  - $\mathcal{O}^z_C$ is $\mathcal{O}_C$ with $U(y') \to U(z)$ (the $y' \to z$ subtraction).

8 Three-ρ rows: two-ρ parts in singlet form

Each block is $\frac{1}{(2\pi)^3}\int \mathcal{K}|_{\rm merge}\,\mathcal{P}\,[\dots]$, to be taken as 2Re.

Group I Row II.

$$\begin{aligned}\frac{d^3N^{(2)}_{\rm II}}{d^3k} = \frac{1}{(2\pi)^3}\int \Big\{ &\mathcal{K}_{\rm II}\big|_{x\to x'}\mathcal{P}(x',y)\Big[- i\,N_c\,{\rm Tr}\big\{\big(U(y)-U(w)\big)U^\dagger(x')\big\} - i\,N_c\,{\rm Tr}\big\{U(w)U^\dagger(w')\big\} \\ &+ i\,{\rm Tr}\big\{T^e U(w')U^\dagger(x')T^e U(y)U^\dagger(w')\big\}\Big] \\ &+ \mathcal{K}_{\rm II}\big|_{y\to x'}\mathcal{P}(x',x)\Big[- i\,N_c\,{\rm Tr}\big\{\big(U(x')-U(w')\big)U^\dagger(w)\big\} + i\,N_c\,{\rm Tr}\big\{U(x)U^\dagger(x')\big\} \\ &- i\,{\rm Tr}\big\{T^e U(w')U^\dagger(x)T^e U(x')U^\dagger(w')\big\}\Big]\Big\}.\end{aligned} \tag{8.1}$$

Group I Row III.

$$\begin{aligned}\frac{d^3N^{(2)}_{\rm III}}{d^3k} = \frac{1}{(2\pi)^3}\int \Big\{ &\mathcal{K}_{\rm III}\big|_{y\to x}\mathcal{P}(x,x')\Big[\tfrac{i}{2}N_c\,{\rm Tr}\big\{\big(U(x')-U(w')\big)U^\dagger(x)\big\} + \tfrac{i}{2}\,{\rm Tr}\big\{T^e U(w')U^\dagger(x)T^e U(z)U^\dagger(w')\big\} \\ &- \tfrac{i}{2}\,{\rm Tr}\big\{T^e U(x)U^\dagger(x')T^e U(x')U^\dagger(z)\big\}\Big] \\ &+ \mathcal{K}_{\rm III}\big|_{x\to x'}\mathcal{P}(x',y)\Big[\tfrac{i}{2}N_c\,{\rm Tr}\big\{\big(U(y)-U(z)\big)U^\dagger(x')\big\} - \tfrac{i}{2}\,{\rm Tr}\big\{\big(U(y)-U(z)\big)U^\dagger(w')T^e U(w')U^\dagger(x')T^e\big\}\Big] \\ &+ \mathcal{K}_{\rm III}\big|_{y\to x'}\mathcal{P}(x',x)\Big[- \tfrac{i}{2}N_c\,{\rm Tr}\big\{U(x)U^\dagger(x')\big\} + \tfrac{i}{2}\,{\rm Tr}\big\{\big(U(x')-U(z)\big)U^\dagger(w')T^e U(w')U^\dagger(x)T^e\big\} \\ &+ \tfrac{i}{2}\,{\rm Tr}\big\{T^e U(x)U^\dagger(x')T^e U(x')U^\dagger(z)\big\}\Big]\Big\}.\end{aligned} \tag{8.2}$$

Group I Row V, part 1.

$$\begin{aligned}\frac{d^3N^{(2)}_{\rm V1}}{d^3k} = \frac{1}{(2\pi)^3}\int \Big\{ &\mathcal{K}_{\rm V1}\big|_{y\to x'}\mathcal{P}(x',y')\Big[i\,N_c\,{\rm Tr}\big\{U(x')U^\dagger(y')\big\} - i\,{\rm Tr}\big\{T^e U(w')U^\dagger(x')T^e U(y')U^\dagger(w')\big\} \\ &+ i\,{\rm Tr}\big\{T^e U(w')U^\dagger(x)T^e U(z)U^\dagger(w')\big\} - i\,{\rm Tr}\big\{T^e U(x)U^\dagger(x')T^e U(x')U^\dagger(z)\big\}\Big] \\ &+ \mathcal{K}_{\rm V1}\big|_{y'\to x'}\mathcal{P}(x',y)\Big[- i\,N_c\,{\rm Tr}\big\{U(x')U^\dagger(y)\big\} + i\,{\rm Tr}\big\{T^e U(w')U^\dagger(x')T^e U(y)U^\dagger(w')\big\} \\ &- i\,{\rm Tr}\big\{T^e U(w')U^\dagger(x)T^e U(z)U^\dagger(w')\big\} + i\,{\rm Tr}\big\{T^e U(x)U^\dagger(x')T^e U(x')U^\dagger(z)\big\}\Big]\Big\}.\end{aligned} \tag{8.3}$$

With Eq. (5.4) the ξ integral of $\mathcal{K}_{\rm V1}$ is in closed form (Ein), and $y \to x'$ or $y' \to x'$ only changes the transverse factor.

$\mathcal{K}_{\rm IV}$ **family** (Group I Row IV, Group III Rows I, II, Group II Row I part 1).

$$\begin{aligned}\frac{d^3N^{(2)}_{\rm IV}}{d^3k} = \frac{1}{(2\pi)^3}\int \Big\{ &\mathcal{K}_{\rm IV}\big|_{y\to x}\mathcal{P}(x,x')\Big[\tfrac{i}{2}N_c\,\mathrm{Tr}\big\{\big(U(x')-U(w')\big)U^\dagger(w)\big\} + \tfrac{i}{2}\,\mathrm{Tr}\big\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\big\}\\ &\quad - \tfrac{i}{2}\,\mathrm{Tr}\big\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\Big]\\ &+\mathcal{K}_{\rm IV}\big|_{x\to x'}\mathcal{P}(x',y)\Big[\tfrac{i}{2}N_c\,\mathrm{Tr}\big\{\big(U(z)-U(w)\big)U^\dagger(x')\big\} + \tfrac{i}{2}N_c\,\mathrm{Tr}\big\{U(w)U^\dagger(w')\big\} - \tfrac{i}{2}\,\mathrm{Tr}\big\{T^eU(w')U^\dagger(x')T^eU(z)U^\dagger(w')\big\}\Big]\\ &+\mathcal{K}_{\rm IV}\big|_{y\to x'}\mathcal{P}(x',x)\Big[-\tfrac{i}{2}N_c\,\mathrm{Tr}\big\{U(w)U^\dagger(x')\big\} - \tfrac{i}{2}N_c\,\mathrm{Tr}\big\{U(x)U^\dagger(x')\big\} + \tfrac{i}{2}\,\mathrm{Tr}\big\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\\ &\quad + \tfrac{i}{2}\,\mathrm{Tr}\big\{T^eU^\dagger(w)U(x')T^eU^\dagger(x')U(z)\big\}\Big]\Big\}. \end{aligned} \tag{8.4}$$

$\mathcal{K}_C$ **family** (Group III Rows III part 1, V, VI). Line 1 and the first subtraction term ($\mathcal{O}_C$):

$$\begin{aligned}\frac{d^3N^{(2)}_{C}}{d^3k} = \frac{1}{(2\pi)^3}\int \Big\{ &\big(\mathcal{K}_C-\mathcal{K}_{C2}\big)\big|_{y\to x}\mathcal{P}(x,x')\Big[-i\,\mathrm{Tr}\big\{\big(U(z)-U(w)\big)U^\dagger(x')T^eU(x')U^\dagger(x)T^e\big\}\\ &\quad + i\,\mathrm{Tr}\big\{\big(U(z)-U(w)\big)U^\dagger(y')T^eU(y')U^\dagger(x)T^e\big\}\Big]\\ &+\big(\mathcal{K}_C-\mathcal{K}_{C2}\big)\big|_{x\to x'}\mathcal{P}(x',y)\Big[-i\,N_c\,\mathrm{Tr}\big\{\big(U(y)-U(w)\big)U^\dagger(x')\big\}\\ &\quad + i\,\mathrm{Tr}\big\{\big(U(x')-U(z)\big)U^\dagger(y')T^eU(y')\big(U^\dagger(y)-U^\dagger(w)\big)T^e\big\} + i\,\mathrm{Tr}\big\{\big(U(y)-U(w)\big)U^\dagger(x')T^eU(x')U^\dagger(z)T^e\big\}\Big]\\ &+\big(\mathcal{K}_C-\mathcal{K}_{C2}\big)\big|_{y\to x'}\mathcal{P}(x',x)\Big[i\,N_c\,\mathrm{Tr}\big\{\big(U(x)-U(z)\big)U^\dagger(x')\big\} - i\,\mathrm{Tr}\big\{\big(U(x)-U(z)\big)U^\dagger(x')T^eU(x')U^\dagger(w)T^e\big\}\\ &\quad - i\,\mathrm{Tr}\big\{\big(U(x)-U(z)\big)U^\dagger(y')T^eU(y')\big(U^\dagger(x')-U^\dagger(w)\big)T^e\big\}\Big]\Big\}. \end{aligned} \tag{8.5}$$

The second subtraction term ($+\mathcal{K}_{C2z}\mathcal{O}^z_C$, the $U(y')\to U(z)$ operator):

$$\begin{aligned}\frac{d^3N^{(2)}_{C,z}}{d^3k} = \frac{1}{(2\pi)^3}\int \Big\{ &\mathcal{K}_{C2z}\big|_{y\to x}\mathcal{P}(x,x')\Big[i\,N_c\,\mathrm{Tr}\big\{U(x)U^\dagger(z)\big\} - i\,\mathrm{Tr}\big\{\big(U(z)-U(w)\big)U^\dagger(x')T^eU(x')U^\dagger(x)T^e\big\}\\ &\quad - i\,\mathrm{Tr}\big\{T^eU(w)U^\dagger(z)T^eU(z)U^\dagger(x)\big\}\Big]\\ &+\mathcal{K}_{C2z}\big|_{x\to x'}\mathcal{P}(x',y)\Big[-i\,N_c\,\mathrm{Tr}\big\{\big(U(y)-U(w)\big)U^\dagger(x')\big\} - i\,N_c\,\mathrm{Tr}\big\{\big(U(y)-U(w)\big)U^\dagger(z)\big\}\\ &\quad + i\,\mathrm{Tr}\big\{\big(U(y)-U(w)\big)U^\dagger(x')T^eU(x')U^\dagger(z)T^e\big\} + i\,\mathrm{Tr}\big\{\big(U(y)-U(w)\big)U^\dagger(z)T^eU(z)U^\dagger(x')T^e\big\}\Big]\\ &+\mathcal{K}_{C2z}\big|_{y\to x'}\mathcal{P}(x',x)\Big[-i\,N_c\,\mathrm{Tr}\big\{U(w)U^\dagger(z)\big\} + i\,N_c\,\mathrm{Tr}\big\{U(x)U^\dagger(x')\big\}\\ &\quad - i\,\mathrm{Tr}\big\{\big(U(x')-U(w)\big)U^\dagger(z)T^eU(z)U^\dagger(x)T^e\big\} - i\,\mathrm{Tr}\big\{\big(U(x)-U(z)\big)U^\dagger(x')T^eU(x')U^\dagger(w)T^e\big\}\Big]\Big\}. \end{aligned} \tag{8.6}$$

9 Two-ρ rows in singlet form

Charges (x',x) or (x',y) as indicated; the kernels are those of Sec. 5. "LO-type" is the colour of LO, the CMW term, DGLAP, the running coupling and $\mathrm{LO}^{\rm lp}_\xi$.

$$\text{LO-type}: \quad \mathcal{P}(x',x)\,\mathrm{Tr}\big\{\big(U(x)-U(y)\big)\big(U^\dagger(x')-U^\dagger(y')\big)\big\}, \tag{9.1}$$

$$H_4: \quad \mathcal{P}(x',x)\,\mathrm{Tr}\big\{\big(U(x)-U(z)\big)\big(U^\dagger(x')-U^\dagger(z)\big)\big\}, \tag{9.2}$$

$$\begin{aligned}\text{G1 V } 2a+2b \text{ (Eq. (5.6), first two lines)}: \quad &\mathcal{P}(x',y)\Big[N_c\,\mathrm{Tr}\big\{\big(U(x')-U(w')\big)U^\dagger(y)\big\} + \mathrm{Tr}\big\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\big\}\\ &\quad - \mathrm{Tr}\big\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\Big], \end{aligned} \tag{9.3}$$

$$\text{G1 V, the } \tfrac{11}{6} \text{ line}: \quad \mathcal{P}(x',y)\,N_c\,\mathrm{Tr}\big\{\big(U(x')-U(w')\big)\big(U^\dagger(y)-U^\dagger(z)\big)\big\}, \tag{9.4}$$

$$\begin{aligned}\text{G2 I part 2 } (2a\text{–}2f, \text{ all}): \quad &\mathcal{P}(x',x)\Big[N_c\,\mathrm{Tr}\big\{U(w)U^\dagger(w')\big\} + N_c\,\mathrm{Tr}\big\{U(x)U^\dagger(x')\big\} - \mathrm{Tr}\big\{T^eU(w)U^\dagger(x')T^eU(z)U^\dagger(w)\big\}\\ &\quad - \mathrm{Tr}\big\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\big\}\Big], \end{aligned} \tag{9.5}$$

$$\begin{aligned}\text{G3 III(2), line 1, } (x'-y') \text{ of line 2}: \quad &\mathcal{P}(x',x)\Big[-N_c\,\mathrm{Tr}\big\{\big(U(x')-U(y')\big)U^\dagger(x)\big\} + \mathrm{Tr}\big\{T^eU(w)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\\ &\quad - \mathrm{Tr}\big\{T^eU(w)U^\dagger(y')T^eU(y')U^\dagger(z)\big\}\Big], \end{aligned} \tag{9.6}$$

$$\begin{aligned}\text{G3 III(2), } (x'-z) \text{ of line 2, line 3}: \quad &\mathcal{P}(x',x)\Big[N_c\,\mathrm{Tr}\big\{\big(U(x)-U(w)\big)U^\dagger(z)\big\} - N_c\,\mathrm{Tr}\big\{U(x)U^\dagger(x')\big\}\\ &\quad + \mathrm{Tr}\big\{T^eU(w)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\Big]. \end{aligned} \tag{9.7}$$

The colour of Group II Row I part 2 is the same ($\mathcal{O}_{58}$) for all of $2a$–$2f$, so that row is $\frac{1}{(2\pi)^3}\int\big(\sum_{2a}^{2f}\text{kernels}\big)\,\mathcal{P}(x',x)[\dots]$.

10 Final result

Prefactors: LO has $\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}$; the NLO kernels have $\mathcal{N} \equiv \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5 k^+} = \frac{\alpha_s^2}{4\pi^6 k^+}$, i.e. $\frac{\alpha_s}{\pi^2}$ relative to LO. For a Gaussian projectile in a colour singlet, $\mathcal{P}(u,v) = \frac{\langle \rho^f(u)\rho^f(v)\rangle}{N_c^2-1}$:

$$\mathrm{LO} = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{P}(x',x)\,\mathrm{Tr}\big\{\big(U(x)-U(y)\big)\big(U^\dagger(x')-U^\dagger(y')\big)\big\}. \tag{10.1}$$

$$\begin{aligned}
\frac{d^3N_h}{d^3k} &= \int_0^1 \frac{d\zeta}{\zeta^3}\,D_{h/g}(\zeta,\mu_F)\,\frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\Big),\\
\frac{d^3N_g}{d^3k} &= \mathrm{LO}\big|_{\alpha_s\to\alpha_s(c_0/|y'-y|)} + \frac{\alpha_s}{2\pi}N_c\Big(\frac{67}{9}-\frac{\pi^2}{3}\Big)\mathrm{LO} + \log\frac{k^+}{\Lambda}\,(\mathrm{JIMWLK}\otimes\mathrm{LO}) + \log\frac{\vee}{k^+}\,(\mathrm{BFKL}\otimes\mathrm{LO})\\
&\quad - \frac{\alpha_s}{2\pi}\int_0^1\frac{d\xi}{\xi^2}\Big\{P_{gg}(\xi)\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big] - 2N_c\frac{1-\xi}{\xi}\mathrm{LO}^y_\xi\big[\ell_\Delta(\mu_F)\big]\Big\}\\
&\quad - H_4 + \frac{\alpha_s}{2\pi}2N_c\int_0^1 d\xi\frac{1-\xi}{\xi}\mathrm{LO}^{\mathrm{lp}}_\xi + \frac{d^3N_{\mathrm{G2\,I(2)}}}{d^3k}\\
&\quad + 2\,\mathrm{Re}\Big[\frac{d^3N^{(2)}_{\mathrm{II}}}{d^3k} + \frac{d^3N^{(2)}_{\mathrm{III}}}{d^3k} + \frac{d^3N^{(2)}_{\mathrm{V1}}}{d^3k} + \frac{d^3N_{\mathrm{V}\,2a+2b}}{d^3k} + \frac{d^3N^{(2)}_{\mathrm{IV}}}{d^3k} + \frac{d^3N^{(2)}_{C}}{d^3k} + \frac{d^3N^{(2)}_{C,z}}{d^3k} + \frac{d^3N_{\mathrm{G3\,III(2)}}}{d^3k}\Big]
\end{aligned} \tag{10.2}$$

Where to find each piece:

term	kernel × colour
H_4, $\mathrm{LO}^{\mathrm{lp}}_\xi$, LO_ξ, LO^y_ξ	Eq. (5.10); colour (9.1) (for H_4: $y,y'\to z$)
G2 I(2)	your $2a$–$2f$ (sum of kernels) $\times\,\mathcal{P}(x',x)\times$ the G2 I part 2 traces of Sec. 9
$d^3N^{(2)}_{\mathrm{II,III,V1,IV,C,C_z}}$	merged kernels (Sec. 7) × traces of Sec. 8
$d^3N_{\mathrm{V}\,2a+2b}$	Eq. (5.6); colour: the $2a+2b$ and $\frac{11}{6}$ traces of Sec. 9
$d^3N_{\mathrm{G3\,III(2)}}$	Eq. (5.14); colour: the two G3 III(2) traces of Sec. 9

Every finite term is $\frac{1}{k^+}\times$ a function of $\mathbf{k}$ only, so it does not depend on rapidity. All of them are bilinear in $\mathcal{P}$. Their colour structures are $N_c\mathrm{Tr}\{UU^\dagger\}$ (dipole) and $\mathrm{Tr}\{T^eAT^eB\}$ (two dipoles joined by a gluon).

11 Comparison with forward rapidity (AABKL, arXiv:1411.2869)

AABKL: a collinear parton of the proton (PDF $x_p f(x_p)$) at forward rapidity, LO $O(\alpha_s^0)$, final result Eq. (4.22) (gluon channel). Ours: a gluon radiated by the charges ρ of a dilute projectile at mid rapidity, LO $O(\alpha_s)$. The dictionary is $x_p \leftrightarrow a = \frac{k^+}{\vee}$; $\vee\to\infty$ is $x_p\to 0$.

AABKL	ours (this limit)	relation
virtual term: $\int_0^1 d\xi\big[\frac{1-\xi}{[\xi]_+}+\frac{\xi}{[1-\xi]_+}+\xi(1-\xi)\big] = -\frac{11}{6}$	G1 V $2a+2b$, Eq. (5.6): weights $\xi(1-\xi)$, $1-\xi$, ξ, $[\frac{1}{\xi}]_+$, $[\frac{1}{1-\xi}]_+$ on $[0,1]$; G1 V(1), Eq. (5.3)	same kernel, same range; our $\frac{11}{6}$ line is their $-\frac{11}{6}$, i.e. $\beta_0 = 2C_A\frac{11}{6}$
real term $\int_0^{1-x_p} d\xi\big[\frac{1-\xi}{[\xi]_+}+\frac{\xi}{1-\xi}+\xi(1-\xi)\big]$, ξ = fraction of the unmeasured gluon	Group III, Eq. (5.13): $\int_a^1 d\zeta$ with $\big[\frac{1}{1-\zeta}\big]_+$ and $\frac{1}{\zeta}$; $1-\zeta$ = fraction of the unmeasured gluon	$\xi_{\mathrm{AABKL}} = 1-\zeta$; upper limit $1-x_p \leftrightarrow \zeta\ge a$ (our $\int_0^{\vee-k^+}dp^+$); $\to\int_0^1$ for $x_p, a\to 0$
shifted dipoles $s\big[y,(1-\xi)\bar y+\xi z\big]$	phase $e^{-i\frac{k}{\zeta}\cdot(y'-[\zeta w+(1-\zeta)z])}$ (G3); $e^{-ik\cdot(w'-[(1-\xi)z+\xi x])}$ (G1 V)	the same weighted centre of the parent; ours in Fourier form
FF term: P_{gg}, $\log\big((y-\bar y)^2\mu^2\big)$, scheme $f_{\mu^2} = f^{\overline{\mathrm{MS}}}_{1.26\mu^2}$	G2 IV, Eq. (5.8), $\ell_\Delta = \log\frac{(y'-y)^2\mu^2}{c_0^2}$	same P_{gg}; $R^2 = c_0^2 = 4e^{-2\gamma_E} = 1.26$
soft term $\int_0^1\frac{d\xi}{\xi}[A_\xi(y-z)-A_\xi(\bar y-z)]^2[\dots]$ (Ioffe-time cutoff; "not an extra evolution")	the subtraction lines: $\frac{1}{\zeta^2}[\dots]_{y'\to z}$ in Eq. (5.13), $\frac{1-\xi}{\xi}\mathrm{LO}^y_\xi$ in Eq. (5.8)	both are what is left of the soft region after the evolution is taken out; scheme dependent (Ioffe time vs. sharp p^+ cut), so not equal term by term
target evolution BK	$\log\frac{k^+}{\Lambda}$ JIMWLK⊗LO	same (BK = large-N_c dipole form)
projectile: PDF DGLAP; no BFKL ($\ln\frac{1}{x_p} = O(1)$)	no PDF DGLAP (emission before the target from the charges is finite, G2 I); BFKL over $\log\frac{\vee}{k^+}$	at mid rapidity $\log\frac{\vee}{k^+} = \frac{1}{2}\log\frac{\vee}{\Lambda}$ is large
no CMW constant (LO $O(\alpha_s^0)$)	$\frac{\alpha_s N_c}{2\pi}(\frac{67}{9}-\frac{\pi^2}{3})$LO	ours renormalizes the coupling of LO
–	three-ρ rows (G1 II, III, IV, V(1); G3 I, II, III(1), V, VI), H_4, linear polarization	only for a projectile of colour charges; a single parton has none

quark channels (C_F)	none ($n_f = 0$)	–

Conclusion. In the limit $\Lambda \to 0$, $\vee \to \infty$ our finite terms have AABKL's structure wherever the physics is the same: the virtual and real $g \to gg$ kernels with the same plus distributions on $[0,1]$, the same limits ($x_p \leftrightarrow \frac{k^+}{\vee}$), the same weighted transverse centres, the same fragmentation DGLAP in the same scheme. They differ where they must: our projectile is a set of colour charges (three-ρ, H_4, linear polarization, finite before-target emission), our LO carries α_s (running coupling and the CMW constant), and the soft remainders are scheme dependent. A numerical term-by-term equality is therefore not expected.

A Exact in $a = k^+/\vee$: where a enters

Your file is exact in a (and verified so). Going from the leading-power forms above to the exact forms needs only these replacements, with $A \equiv (x-z)^2$, $B \equiv (x-w)^2$, $\mathcal{L}_a(x;z,w) \equiv \log \frac{(1-a)(x-z)^2 + a(x-w)^2}{(x-w)^2}$:

term	exact in a
hadron, DGLAP, $\mathrm{LO}^{\mathrm{lp}}$	$\int_{k^+/\vee}^1 \frac{d\zeta}{\zeta^3}$; $\int_a^1 \frac{d\xi}{\xi^2}$ with the standard plus distribution on $[a,1]$ (E4); $\int_a^1 d\xi \frac{1-\xi}{\xi} \mathrm{LO}^{\mathrm{lp}}_\xi$; $-H_4 \to -(1-a)H_4$
G1 II part 1	+Q1 and $V_\vee$, Sec. 3
G1 II part 2	R^i of your file ($\log \frac{p^2}{k^2+ap^2}$, $\log\left[\frac{k^2}{(k-p)^2} + a\left(1 - \frac{k^2}{(k-p)^2}\right)\right]$, $-\frac{p^2k^i + k^2p^i}{(k-p)^2}\log(1-a)$)
G1 III	part 1: $\log\left\lvert\frac{A-aB}{A-B}\right\rvert$, $\log\left\lvert\frac{A-aB}{B}\right\rvert$ and a $\log(1-a)$ line; part 2: $\log\left\lvert\frac{aB-A}{B-A}\right\rvert$ and a $\log(1-a)$ line (your equations)
$\mathcal{K}_{\mathrm{IV}}$ family (G1 IV, G2 I part 1, G3 I, II)	$\log\frac{A}{B} \to \mathcal{L}_a(x;z,w)$, and $+\log(1-a)\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2} + \frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]$ inside the bracket of $\mathcal{K}_{\mathrm{IV}}$ (same weights $1, -\frac{1}{2}, -1, -1$)
G2 I part 2	$\log\frac{(x-z)^2}{(x-w)^2} \to \mathcal{L}_a(x;z,w)$, primed likewise; $2a$: $\log\frac{(1-a)A+aB}{a(x'-w')^2+(1-a)(x'-z)^2}$; $2b$: a $\log(1-a)$ line; $2f$: $-\frac{1}{k^+} \to \frac{1}{\vee} - \frac{1}{k^+}$
Group III	$\int_a^1 d\zeta$ (your $\int_0^{\vee - k^+} dp^+$); your $\log(1-a)$ lines are exactly the endpoint term of the standard plus distribution: $\int_0^{\vee-k^+} dp^+ \frac{k^+[F(p^+)-F(0)]}{p^+(p^++k^+)} + \log(1-a)F(0) = \int_a^1 d\zeta \left[\frac{1}{1-\zeta}\right]_+ G(\zeta)$ (checked, 2×10^{-16}); G3 III(2) third line: $\log\frac{B}{A} \to \log\frac{B}{(1-a)A+aB}$ (checked)

The colour factors, hence all traces, do not depend on a.

B Open points for the paper

1. **Four-ρ finite term.** Your files have none: `4rho.tex` has only $\left[\log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+}\right]$ terms. For a Gaussian projectile a finite four-ρ term would give $\mathcal{P}\mathcal{P}$ terms. State in the paper that there is none, if that is what the calculation gives.
2. **R^i at large $|x-y|$.** The $p \to 0$ and $k-p \to 0$ ends of $\mathcal{K}_{\mathrm{II}}$ give $\int d^2p \frac{\log p^2}{p^2}$-type behaviour, i.e. large $|x-y|$ (`uv_check`, F5). It is regulated by the finite size of the projectile (the range of $\mathcal{P}$), not by μ; it is not a UV or collinear divergence.
3. **Beyond Gaussian.** The symmetric three-ρ parts $S(\mathcal{O})$ vanish only for a Gaussian. For a general colour-singlet projectile they are $\propto d^{abc}\langle d\rho\rho\rho\rangle$, $f^{abc}\langle fS(\rho\rho\rho)\rangle$ (`finite_singlet`).
4. **Exact a**: only with the virtual ξ integral on $[0,1]$ (Q1).
5. **The constant $2K$** (Eq. (3.3)): keep it explicit; it is twice the CMW shift.

C Checks

check	result (file)
25 of 30 equations identical to the verified ones	`checks.json`
$2b$ UV in the form of G1 II(1): β_0 term $-\frac{11}{3}$ (cancels $+\frac{11}{3}$); $\log(1-\frac{\Lambda}{k^+})$ coefficient 4	`checks.json` (sympy)
leftover pole: $-N_c\log(1-a)$ (Q1)	`checks.json`
Group III: plus prescription $+\log(1-a)$ line = standard plus on $[a,1]$; $\frac{dp^+}{p^++k^+} = \frac{d\zeta}{\zeta}$ on $[a,1]$	2×10^{-16}, 6×10^{-17}
G3 III(2) closed form, exact a	2×10^{-16}
$\mathcal{N} = \frac{\alpha_s^2}{4\pi^6 k^+}$, NLO/LO $= \frac{\alpha_s}{\pi^2}$	exact
exact-a vs $a \to 0$, all equations	$\propto a$ (`limiting/limcheck.json`)
Group III large p^+	$\sim 1/p^{+2}$ (`limiting/largep.json`)
simplifications (Ein, ζ map, $q^+ \to \xi$, phases, $2a$ colour at $z=x$)	$\le 2\times10^{-15}$ (`limiting/checks.json`)
traces vs direct contraction; pair rule vs engine; $U(y') \to U(z)$ operators	$\le 8\times10^{-16}$, $\le 2.4\times10^{-16}$, exact (`limiting/singlet3.json`)
G2 IV: DGLAP, RC, line 2 has no μ	`g2iv_rc/compact/checks.py`